

Rest Assured: The SQL Side of Sleep Quality

When I first started my project, I had no idea just how intertwined our daily habits are with our sleep. One late night, while I was tossing and turning, it hit me—what if there's a way to understand the factors that impact our sleep? This thought sparked my investigation into the various health and lifestyle factors that contribute to getting quality sleep.

Why THIS Project?

As someone who struggles with sleep regularly, I was motivated to explore what truly influences the quality and duration of our rest. I wanted to find concrete answers to how physical health and lifestyle choices affect our sleep. This project felt personal; it wasn't just data to me—it was a journey to uncover insights that could make a difference in how we approach our sleep habits.

What Readers Will Gain

By diving into my findings, you'll learn about:

- The surprising connections between your weight and sleep quality.
- How stress impacts your ability to rest well.
- Insights on how different occupations relate to sleep patterns.

Key Takeaways

- Engineers generally achieve better sleep, regardless of their activity levels.
- Individuals who are overweight or obese tend to sleep less than those of normal weight.
- Lower stress levels lead to improved sleep quality.

Dataset Details

I used a dataset from Kaggle, which contains 374 records. This data was perfect for my exploration as it covered various aspects of health and lifestyle, providing a broad perspective on sleep quality.

Analysis Process

I began by cleaning the data to ensure accuracy. This included removing any duplicates and outliers that could skew my results. Then, I explored relationships within the data through visualization techniques. I used charts to display sleep quality against physical activity and stress levels, which helped me identify trends.

What surprised me was the significant impact of BMI on sleep duration. I discovered that higher BMI often correlates with lower sleep duration, which is something I hadn't fully realized before.

Visuals and Insights

1. **Occupation and Sleep Quality Chart:** This chart presents the physical activity levels, average sleep quality, and stress level by occupation. Interestingly, it shows that engineers consistently report better sleep quality, even when they maintain varying levels of physical activity.

```
SELECT
    Occupation,
    "Physical Activity Level",
    ROUND(AVG("Quality of Sleep"), 2) AS Avg_Sleep_Level,
    ROUND(AVG("Stress Level"), 2) AS Avg_Stress_Level,
    COUNT(*) AS Sample_Size
FROM
    "Sleep and Health"
GROUP BY
    Occupation, "Physical Activity Level"
HAVING
    COUNT(*) > 5
ORDER BY
    Avg_Sleep_Level DESC, Avg_Stress_Level ASC;
```

	Occupation	Physical Activity Le...	Avg_Sleep_Quality	Avg_Stress_Level	Sample_Size
1	Engineer	30	9	3	32
2	Nurse	75	8.97	3.09	34
3	Accountant	60	8	4	26
4	Engineer	60	8	4	6
5	Teacher	60	8	4	6
6	Lawyer	90	8	5	17
7	Engineer	90	8	5	16
8	Lawyer	60	7.93	5	27
9	Doctor	75	7	6	32
10	Accountant	45	7	7	6
11	Teacher	45	6.93	4.22	27
12	Salesperson	45	6	7	32
13	Doctor	30	6	8	31
14	Nurse	90	6	8	32

2. **BMI and Sleep Duration Chart:** The second chart compares categories of BMI with average heart rates and sleep duration. It clearly indicates that those classified as obese experience higher heart rates and shorter sleep durations compared to those with lower BMI. This pattern highlights the importance of weight management for better sleep quality.

```

SELECT
  CASE
    WHEN "BMI Category" IN ('Normal', 'Normal Weight') THEN 'Normal
Weight'
    ELSE "BMI Category"
  END AS Cleaned_BMI_Category,
  COUNT(*) AS Sample_Size,
  ROUND(AVG("Heart Rate"), 2) AS Avg_Heart_Rate,
  CASE
    WHEN AVG("Heart Rate") < 70 THEN 'Low'
    WHEN AVG("Heart Rate") BETWEEN 70 AND 80 THEN 'Normal'
    ELSE 'High'
  END AS Heart_Rate_Status,
  ROUND(AVG("Sleep Duration"), 2) AS Avg_Sleep_Duration
FROM
  "Sleep and Health"
GROUP BY
  Cleaned_BMI_Category
ORDER BY
  Avg_Heart_Rate DESC;

```

	Cleaned_BMI_Cate...	Sample_Size	Avg_Heart_Rate	Heart_Rate_Status	Avg_Sleep_Duration
1	Obese	10	84.3	High	6.96
2	Overweight	148	70.95	Normal	6.77
3	Normal Weight	216	68.98	Low	7.39

Main Takeaways

From my analysis, it's evident that stress levels, physical activity, and weight play significant roles in determining how well we sleep. These findings can lead to actionable steps for individuals seeking restorative sleep. For instance, managing stress through mindfulness practices or regular exercise could lead to improved sleep quality.

Conclusion and Personal Reflections

Throughout this project, I learned a lot about the correlations between our lifestyle choices and sleep quality. One of the biggest challenges I faced was ensuring the accuracy of the data; however, this pushed me to be more thorough and detail oriented. I've come away with a deeper understanding of how factors like stress and lifestyle choices can impact our health and well-being, shaping my goals for the future.

Call to Action

I invite you to connect with me on LinkedIn! If you or anyone you know is looking to hire a data analyst, let's chat. Your thoughts and feedback are welcome, too!